A Comparison of Cross-company and Within-company Effort Estimation Models for Web Applications

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Abstract

The objective *of this paper is to investigate* ***t****he use of cross-company and within-company cost estimation model for Web projects.*

The study analysis method *we used was a* ***f****orward, stepwise regression of web project development effort with independent variables that were used for on-line web application price quotations. Model cross-validation used two approaches: one-fold cross validation and a validation subset of 13 projects from the same company.*

The setting consists of *Industrial Web applications from 24 different companies and 8 different countries, and the experiment units are 53 web application and hypermedia projects.*

The main outcome measure *was an estimate of effort for each project, for each estimation model. We used the summary statistics: MMRE, Median MRE, Pred(25), mean and median of absolute residuals to evaluate the predictive accuracy of model. We also compared estimation model accuracy with an assessment of estimate accuracy provided by the participating companies.*

Results showed that *the best fitting cross-company regression model was significantly better than the median and experts at predicting effort for 13 projects from a single company ($p<0.05$). The best fitting within-company regression model for the company with 13 projects was significantly better than the cross-company model ($p<0.05$). However the accuracy of the company 1 model was worse than the accuracy achieved by company 1 estimators. In addition, the company 1 model was an extremely poor predictor of effort for the other companies' projects.*

We conclude that *cross-company effort estimation model can be useful for companies that do not have past projects from which to develop their own models. However, a cross-company data set should only be used for estimation until it is possible for a Web company to build its own within-company model. For cross-company and within company models, simple measures, available early in the lifecycle, that can be estimated by clients are suitable for building effort estimation models.*

Keywords: effort estimation, Web projects, cross-company estimation models, within-company estimation model, regression-based estimation models.

1. Introduction

There is considerable interest in using cross-company data sets to construct a company specific estimation model in order to overcome the difficulty of accumulating sufficient in-house data (see for example [11], [13]). Indeed most benchmarking companies work on the assumption that it is possible to construct good cross-company models. However, support for this assumption is mixed. Early evaluations of cost models suggested that it was not possible to take a model built in one environment, and apply it without calibration in another environment ([7],[9]). However recently there have been four major studies of large cost estimation data sets ([1],[2],[5],[6]). These studies have used data from two application domains: 'business' and 'space and military'. In two studies ([1],[2]) the researchers found that a cross-company model worked well when applied to a specific company, but in the other two studies ([5],[6]) they did not observe the same effect. The two cases where the data were collected using rigorous quality-assurance procedures found cross-company models suitable for within-company predictions, whereas the two cases that did not use rigorous data collection procedures did not find such models. In addition, Wieczorek and Ruhe reanalysed one of the data sets and found that within-company models were no better than cross-company models [17]. None of these studies used data on Web projects.

This paper reports on the analysis of a cross-company data set of Web projects. We have constructed and evaluated a cost estimation model based on this data to investigate how good a cross-company data set is:

- In its ability to predict effort for a single company (omitting projects from that company from model construction).
- In comparison with a company-specific model.

In addition, we also checked the extent to which a company-specific cost model can be used to predict effort for other companies' projects.

Throughout this paper a Web project can either represent a Web hypermedia or Web software application [3]. The former is characterised by the authoring of information using nodes (chunks of information), links (relations between nodes), anchors, access structures (for navigation) and its delivery over the Web. Technologies commonly used for developing such applications are HTML, JavaScript and multimedia. In addition, typical developers are writers, artists and organisations that wish to publish information on the Web and/or CD-ROMs without the need to use programming languages such as Java. Conversely, the latter represents software applications that depend on the Web or use the Web's infrastructure for execution. Typical applications include legacy information systems such as databases, booking systems, knowledge bases etc. Many e-commerce applications fall into this category. Typically they employ development technologies (e.g., DCOM, ActiveX etc), database systems, and development solutions (e.g. J2EE). Typical developers are young programmers fresh from a Computer Science or Software Engineering degree, managed by more senior staff.

The remainder of the paper is organised as follows: Section 2 describes the research method employed in this study and results are presented in Section 3. Finally, a discussion of the results and conclusions is given in Section 4.

2. Research Method

2.1 Data set Description

The analysis presented in this paper was based on Web projects from the Tukutuku database [14]. These projects represent industrial Web applications developed by Web companies worldwide. This database is part of the Tukutuku project, which aims to collect data about Web projects, to be used to develop Web cost estimation models and to benchmark productivity across and within Web Companies[1].

The analysis presented in this paper used data from 53 Web projects where 13 projects come from the same Web company and the remaining 40 come from another 23 companies. Each Web project in the database provided 40 variables to characterise a Web application and its development process (see Table 1).

The independent variables were selected as a result of a survey investigation [14], based on data from Web application pricing forms gathered worldwide. The variables included early Web size measures and other cost drivers. In addition, the variables were validated by a mature Web company [14]. Consequently we believe we have used variables for prediction that are meaningful and realistic measures both for Web companies and their clients.

Table 1 Variables for the Tukutuku database

Variable Name	Scale	Description
Country	Nominal	Country company belongs to
Established	Ordinal	Amount of time company has been established
Services	Nominal	Services Company provides
ClientInd	Nominal	Industries representative of clients
TypeProj	Nominal	Type of project
TypeApp	Nominal	Type of Web application developed
Totwp	Ratio	Number of web pages
Nnp	Ratio	Number of New Web pages
WPC	Ratio	Web pages given by the customer
WPO	Ratio	Web pages developed by third party
Textyp	Ratio	Number text pages typed (~600 words)
Textel	Ratio	Number text pages electronic format
TSC	Ratio	Number text pages scanned
Imanew	Ratio	Number new images
I3P	Ratio	Number images developed by third party
ISC	Ratio	Number images scanned
ILB	Ratio	Number images reused from a library
Aninew	Ratio	Number new animations
ALB	Ratio	Number animations reused from a library
AVN	Ratio	Number new audio/video files
AVL	Ratio	Number reused audio/video files
Procs	Ratio	Number <> products application offers
HED	Ratio	Development Effort corresponding to high effort[2]
HEA	Ratio	Adaptation/Modification Effort corresponding to high effort
HFots	Ratio	Number reused High effort features/functions without adaptation
HFotsA	Ratio	Number adapted High effort features/functions

[1] http://www.cs.auckland.ac.nz/Tukutuku/introduction.html

[2] High effort is taken to be on average 12 hours for a new feature and 8 hours for an adapted feature.

Newhf	Ratio	Number new High effort features/functions
Tothf	Ratio	Total Number high effort features/functions
Fots	Ratio	Low effort FOTS
FotsA	Ratio	Low effort FOTS-A
Newlo	Ratio	Number new Low effort features/functions
Totlo	Ratio	Total Number low effort features/functions
EFF	Ratio	Total effort develop the Web application
EFA	Nominal	Procedure used to record effort data
Languages	Nominal	Implementation languages used
DocProc?	Nominal	Project followed defined and documented process
ProcImpr?	Nominal	Development team involved in a process improvement programme
Metrics?	Nominal	Development team part of a software metrics programme
Devteam	Ratio	Size development team
Teamexp	Ratio	Average team experience with the development language(s) employed

2.2 Data Quality

We did not supply Web companies that provided data for the Tukutuku database with automatic measurement tools. Instead, we relied on their own data collection mechanisms, so we have no means of assessing the accuracy of their data. In order to have a better idea of how reliable the effort data was, we asked companies how their effort data was collected (see Table 2).

Table 2 How effort data was collected

effort data	Scale	Number of Projects and Companies	
		# and % projects	# different companies
No timesheets	1	12 (22.6%)	8
Total hours worked each day or week	2	3 (5.7%)	3
Hours worked per project per day/week	3	24 (45.3%)	12
Hours worked per project task per day	4	14 (26.4%)	7

Two companies used different data collection levels depending on the type of project (i.e. they used level 1 for some projects and levels 3 and 4 for other projects). All 13 projects from the same company recorded hours worked for each project for each day/week. Therefore, at least for 72% of Web projects in the Tukutuku database effort values were based on more than post-hoc guesses. However, even when timesheets are in use we cannot assume that effort values are completely accurate.

The data collected to date for the Tukutuku project has not followed rigorous quality assurance procedures to validate the data. Furthermore, the projects' application domains are mixed.

2.3 Modelling Techniques

We used the STATA™ tool to perform all data analysis reported in section 3.

A manual forward stepwise regression was employed to build an effort prediction model using the variables shown in Table 3. This is similar to the stepwise technique suggested by Kitchenham [8] but we used the additional variable plot facility in the STATA tool to assess the impact of each candidate variable on the residuals from the current model in each step.

Table 3. Variables used in the stepwise regression

Variable	Meaning
lneff	Natural log. of the company effort to produce the web application
lntotwp	Natural log. of number web pages in the application
lnnp	Natural log. of (1+number of new web pages)
lntothf	Natural log. of (1+total number of high effort functions)
lnnewhf	Natural log. of (1+number of new high effort functions)
lntotlo	Natural log. of (1+ total number of low effort functions)
lnnewlo	Natural log. of (1+number of new low effort functions)
lnprocs	Natural log. of (1+number of different products the web application offers)
lnaninew	Natural log. of (1+number of new animations in the applications)
lnimanew	Natural log. of (1+number of new images in the applications)
lntextyp	Natural log. of (1+number of text pages typed)
lntextel	Natural log. of (1+number of text pages already in electronic format)
devteam	The number of people who worked on the project
teamexp	Average team experience with the development language(s) employed
Natural log. = Natural logarithm	

The set of variables used for model building was a subset of the full Tutukutu database. Variables were excluded if most instances were zero, or they were functionally related to other variables (e.g. number of reused pages was ignored because it is functionally related to the variables total web pages and new web pages) or they were categorical variables. The dependent variable (Total effort) was transformed to a natural logarithmic scale to stabilise the variance and reduce the impact of outliers. Highly skewed independent variables were also subjected to a natural logarithmic

transformation. Note, to cope with zero values for variables, 1 was added to variables that had zero values prior to the log transformation.

2.4 Analysis Methods

Model Evaluation: Stability

We used the residual analysis facilities available in the STATA tool to identify large residual and high influence data points (i.e. projects). In particular we used Cook's D statistic to identify projects that exhibited jointly a large influence and large residual [4]. The stability of the model was tested by removing the high-influence, large-residual projects, and observing the effect their removal had on the model parameters.

Model Evaluation: Goodness of fit

The model's goodness of fit model was tested on the raw data scale using the MMRE and Pred(25) statistics, the median and mean absolute residual [10] and an assessment of their current estimate accuracy provided by ten of the participating companies. This assessment was based on an educated guess.

Prediction Accuracy

Prediction accuracy of models was checked either by using the 'omit one project at a time' procedure (leave one out cross-validation), or by omitting a group of projects and predicting the effort for the group of omitted projects. In both situations, the rationale is to use different sets of projects to build and to validate a model. Projects used to build a model are called the training set and projects used to validate the model are called the validation set.

3. Results

3.1 Model Construction

The best-fitting model for log total effort included three variables: log total number of web pages, log (1 + total number of high effort functions), and the size of the development team (see Table 4). Its adjusted R^2 was 0.597. This model was based on the full set of 53 Web projects.

The best fitting model can be transformed back to the raw data scale as follows:

$$Eff = 8.42497 \times (Totwp)^{0.4332} \times (Tothf+1)^{0.6329} \times e^{0.2354 \times Devteam}$$

Table 4 Best Fitting Model (dependent variable lneff)

Indep. Vars	Coeff.	Std. Error	t	p>|t|	95% Conf. Interval
lntotwp	0.4332	0.1025	4.224	0.000	0.227-0.6392
lntothf	0.6329	0.1859	3.405	0.001	0.2594-1.0064
devteam	0.2354	0.0955	2.464	0.017	0.0434-0.4272
constant	2.1312	0.4029	5.290	0.000	1.3219-2.9412
Indep. Vars – independent variables Coeff. - Coefficient					

3.2 Model Evaluation

Before accepting the model for lneff presented in Section 3.1, we need to make sure that [12]:

Step 1. The errors in our model are random and normally distributed. This is investigated using a residual plot showing residuals vs. fitted values.

Step 2. No single project or small number of projects has an overly strong influence on the results. Here we need to use a leverage plot and to calculate Cook's D values for all projects, in order to detect influential cases.

Step 1: Testing the Residuals

The residual plot (see Figure 1) shows that projects 21, 16, and 17 have large residuals that are far from the other projects. However, overall the ellipsoid pattern of the residuals suggests that they are approximately normally distributed.

Step 2: Leverage and Stability Analyses

The leverage plot (Figure 2) identified several projects with large residuals and high influence, shown encircled. This was investigated further by calculating Cook's D values (i.e. values greater than 4/n where n=53) for each project. The Cook's D statistic attempts to convert the information in the leverage diagram into a single statistic.

Table 5 identifies the projects with significant Cook's D values, showing some of their variable values.

Table 5 Projects with large Cook's D values

Pro	te	lneff	Pred	lntotwp	lntothf	dv
14	90	4.4998	5.5511	1.791759	1.94591	6
17	6	1.7918	4.1386	2.484907	1.098612	1
20	625	6.4378	4.6058	2.995732	0	5
33	3150	8.0552	6.2730	3.091043	2.197225	6
16	363	5.8944	8.3026	6.39693	2.397895	8
Pro – project te – toteffor Pred – prediction dv - devteam						

In order to check the stability of the model, we recalculated the multiple regression omitting the projects with large Cook's D values, giving the model shown in Table 6. The adjusted R^2 for this model is 0.715. The parameters in this model have similar values to those in the model based on the full data set. In addition, the

model fit has improved. This implies that the high influence data points are not destabilising the regression model and therefore do not need to be removed.

Figure 1 – Residual plot for best-fitting model (with points indicated by the project number)

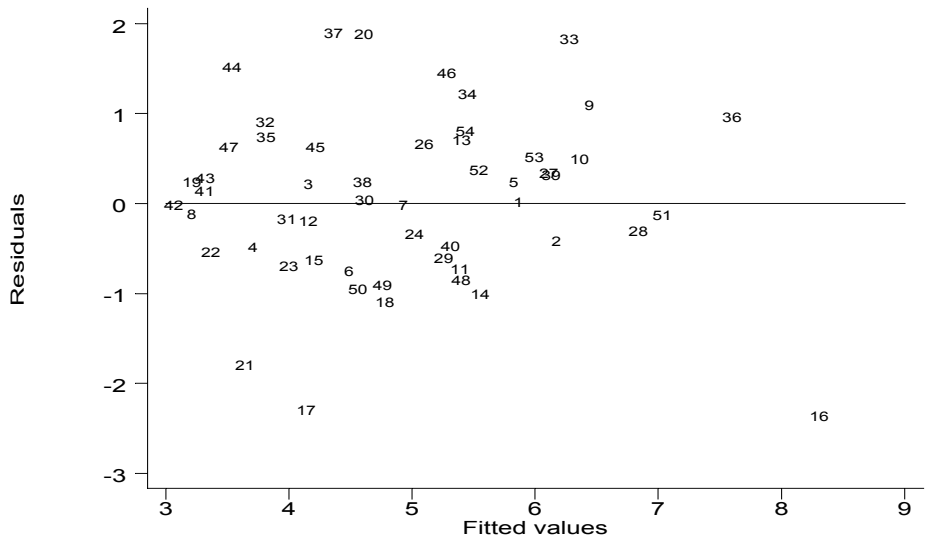


Figure 2 Leverage plot for best-fitting model (with points indicated by the project number)

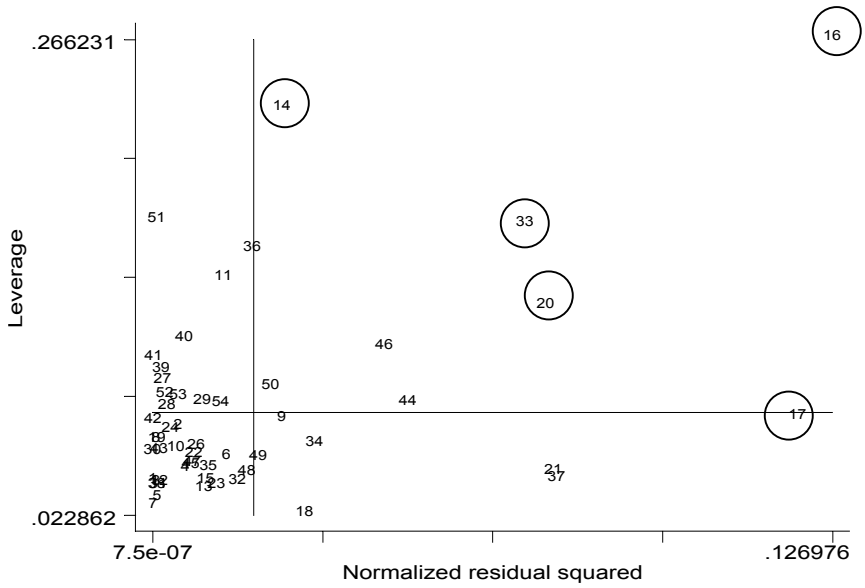


Table 6 Regression model with high influence projects omitted (lneff as dependent variable)

ind	Coef.	Std. Error	t	p>|t|	[95% Conf. Interv.]	
lntotwp	0.4171	0.0936	4.458	0.000	0.2286	0.6059
lntothf	0.7915	0.1604	4.935	0.000	0.4682	1.1147
devteam	0.2590	0.1077	2.406	0.020	0.0420	0.4760
_cons	2.0647	0.3503	5.895	0.000	1.3588	2.7706
ind - Independent variables						

3.3 Goodness of Fit and Model Value

Although the model looks a reasonable fit on the log scale, it is important to remember that any estimates must apply to the raw scale. The standard goodness of fit statistics for the regression model (after transformation

back to the raw data scale) are shown in Table 7. They show that the model is a poor fit to the data. The difference between the mean and median MRE and the mean and median absolute deviation indicate that the poor performance is due to the impact of outliers. In addition, MMRE is worse than the estimate accuracy of a 68.3% underestimate provided by the participating companies (excluding company 1). Actual and estimated effort and size for the worst estimated projects on the raw scale are shown in Table 8.

Table 7 Goodness of fit statistics for the regression model and the median model

Goodness of fit statistic	Estimates based on regression model
MMRE	1.05
Median MRE	0.534
Pred(25)	26.4%
Mean absolute deviation	307.0
Median absolute deviation	71.9
Estimate accuracy from expert opinion	68.3%

Table 8 Effort and Size of badly estimated projects

Project	Effort Estimate	Effort	Web pages
9	618.8	1786	100
16	4059.8	363	600
33	522.4	3150	22
36	2041.6	5000	1390

3.4 Comparison of Cross-company and Within-company Models

In this Section we investigate the issues of how good a cross-company data set is:

- In its ability to predict effort for a single company (omitting projects from that company from model construction).
- In comparison with a company-specific model.

In order to evaluate the ability of the cross-company model to predict effort for a single company we recalculated the model presented in Section 3.1, after excluding all the Company 1 projects (i.e. all the 13 projects provided by a single company). (Note, we did not perform a separate stepwise regression, we simply performed a regression using the variables shown in Table 4 omitting the 13 projects to be predicted). We then used Company 1 projects as our validation set, and calculated MRE and residuals for each. We also compared the predictions for Company 1 with a prediction based on the median of the data set (where the median project effort excluding Company 1 projects is 103.5 hours). The box plots of the residuals are shown in Figure 3.

The Wilcoxon matched-paired signed rank tests, based on absolute residuals, confirmed that the regression predictions were significantly better than the predictions from the median model ($p<0.05$). This was also confirmed by the paired 't' test ($p<0.05$).

Figure 3 Box plot of regression-based predictions compared with median predictions for the 13 Company 1 projects

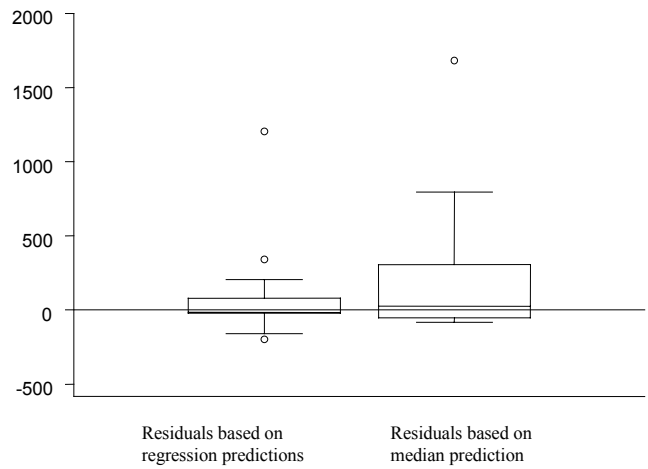


Figure 4 Scatter plot of log effort against log total web pages showing Company 1 projects (labelled 1) and other projects (labelled 0).

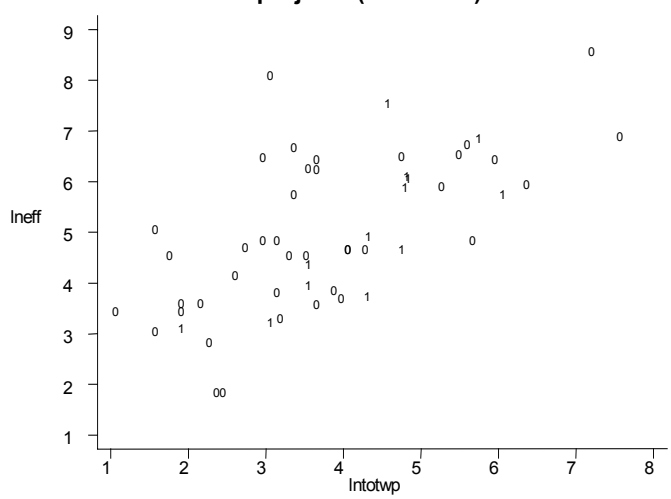


Table 9 shows the prediction statistics for the regression based predictions and the median based predictions. They confirm that the regression based predictions are better than the median based predictions and better than the estimate accuracy of a 68.3% underestimate provided by the participating companies (excluding company 1). Unexpectedly, the regression based prediction statistics given in Table 9 are better than the goodness of fit statistics for the regression model shown in Table 7. This may be because the Company 1 projects are particularly well-behaved (see Figure 4 which

shows how the Company 1 projects are distributed among the other projects). However, it is still remarkable how well the cross-company model fits the Company 1 projects, particularly since the best-fitting model for Company 1 is somewhat different to the best fitting model for the entire data set (see Table 10).

The best fitting model for Company 1 is shown in Table 10. Instead of including the logarithm of the total number of web pages as the most important variable, it includes team experience as the least important variable with a negative effect. The adjusted R^2 for this model is 0.966. This is a suspiciously large value. It is primarily due to the strong correlation between the natural logarithm of effort and the natural logarithm of total number of high effort features shown in Figure 5. Note, the number of zero values for the dependent variable is a cause for concern. The model may well be a poor prediction model if future projects do not include high effort features.

Table 9 Prediction statistics for the regression and median models

Prediction statistics	Predictions based on regression model	Predictions based on median effort
MMRE	0.565	1.169
Median MRE	0.444	0.759
Pred(25)	30.8% (4 of 13)	15.4% (2 of 13)
Mean absolute deviation	179.5	298.6
Median absolute deviation	77.5	79.5

Table 10 Regression model for Company 1 (lneff as dependent variable)

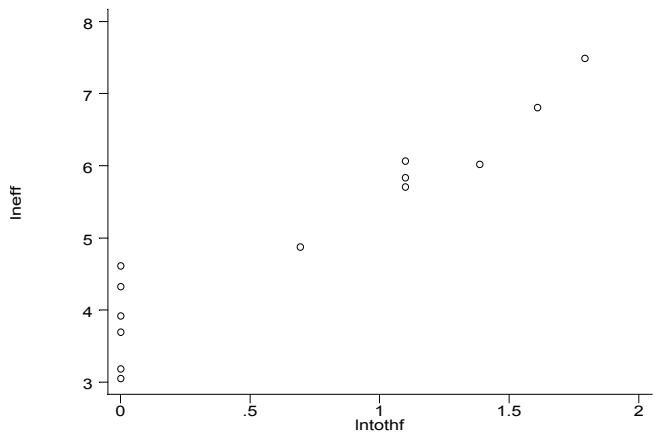
Independent variables	Coef.	Std. Err.	t	P>|t|	[95% Conf. Interval]	
lntothf	1.5510	0.1273	12.184	0.000	1.2630	1.8390
devteam	0.2611	0.0661	3.949	0.003	0.1116	0.4107
teamexp	-0.0881	0.0332	-2.650	0.026	-0.1632	-0.0129
_cons	3.6214	0.2343	15.455	0.000	3.0913	4.1514

The best fitting model can be transformed back to the raw data scale as follows:

$$\text{Eff} = 37.3898 \times (\text{Tothf}+1)^{1.5510} \times e^{0.2611 \times \text{Devteam}} \times e^{-0.0881 \times \text{Teamexp}}$$

In order to evaluate the predictive accuracy of the cross-company model in comparison with a company-specific model, we used the company-specific model (see Table 10) and constructed predictions for Company 1 projects using the 'omit one project at time' procedure (i.e. each of the 13 validation sets had only one project). (Note, we did not perform a separate stepwise regression for each prediction, we simply performed a regression using the variables shown in Table 10 omitting the project to be predicted). The results of this procedure are shown in Figure 6, where the residuals have been transformed to the raw data scale. Both the Wilcoxon matched-paired signed rank test and the paired 't' test confirm that the absolute residuals for the Company 1 model are significantly smaller than the absolute residuals for the model based on the other companies. The summary statistics for the Company 1 model are shown in Table 11, which confirms that the within-company model is much superior to the cross-company model. The estimate accuracy achieved by Company 1 was a 10% underestimate, which is superior to the accuracy provided by our model. This is an extremely good accuracy and is not representative of the accuracy achieved by other companies in our dataset. The only other report of Web project estimate accuracy that we are aware of reported an MMRE of 0.37 for a small Web company that usually underestimated effort [16].

Figure 5 Scatter plot of effort against high effort functions (natural log-log scale)



To check the extent to which a company-specific cost model can be used to predict effort for other companies' projects we used the Company 1 model to predict the values of the other companies' projects. Unfortunately, Table 12 shows that the Company 1 model is a very poor predictor of effort for the other projects.

Table 11 Summary statistics for predictions

Prediction statistics	Predictions based on Company 1 model	Predictions based on other company model
MMRE	0.245	0.565
Median MRE	0.234	0.444
Pred(25)	53.8% (7 of 13)	30.8% (4 of 13)
Mean absolute deviation	61.3	179.5
Median absolute deviation	35.9	77.5

Figure 6 Residuals for Company 1 based on the Company 1 model and the Model based on other companies

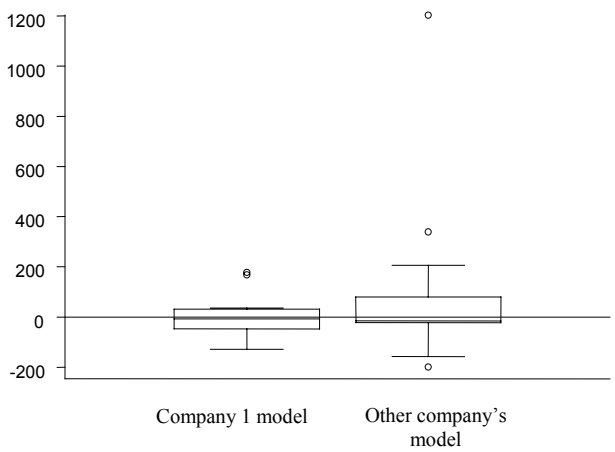


Table 12 Predictions of the other Companies' projects based on Company 1 model

Prediction Statistics	Prediction for other companies based on Company 1 model
MMRE	3.57
Median MRE	0.78
Pred(25)	15%
Mean absolute deviation	632.1
Median absolute deviation	182.2

4. Discussion

The results reported in this paper refer to a specific data set. The data was not collected by means of a randomised survey, so the conclusions that can be drawn from the data are limited. In particular, the specific models presented in the paper cannot be assumed to apply to companies other than those who contributed data, or, perhaps, to companies that develop applications similar in size and type to those described here. We are currently collecting data from other companies, which will allow us to assess whether the cross-company model can be applied more widely. Thus, that the main scientific contribution of our paper is to provide examples (or counter-examples) of cost estimation phenomena in the Web development scenario that can be compared with results from other studies.

Our results support the view that it is possible to construct a cross-company effort estimation model. In addition, it is interesting to observe that two of the three variables included in the best-fitting model (total number of web pages and total number of high effort functions) were the two top ranking measures used by Web companies in their own pricing models [14].

Although the cross-company regression model (see Table 4) was statistically significant, it only accounted for 60% of the variation on the transformed scale and exhibited poor goodness of fit statistics on the raw data scale (MMRE=1.05 and median MRE=0.534). However, the model behaved better when used to predict Company 1 projects (MMRE=0.565 and Median MRE=0.444). In both cases our models presented better estimation accuracy than the estimate accuracy achieved by the participating companies. This result supports two of the previous studies that have looked at the ability of cross-company models to predict individual company results. However, in contrast to the previous studies, our results were found for a data set where it was not possible to apply rigorous quality control procedures to the data collection process.

In contrast to [17], our results also show a substantial improvement if a company specific model is used. For our data set, this suggests that a cross-company data set should only be used for estimation until it is possible to build a within-company model. Furthermore, our within-company model included different variables than the cross-company model. This implies that simple recalibration of an existing cross-company model will not necessarily lead to the best model. However, our within-company model was no use at all for predicting the effort of the projects in other companies. This is consistent with the early studies of cost estimation models which suggested models could not be used outside the environment in which they were developed.

Further study using Web projects data is necessary to confirm our findings. At the time this study was conducted only one company provided sufficient data to construct a within company model.

In summary, for practitioners who work with Web development and manage Web projects, here are the key observations we believe can be gained from our results:

- A cross-company effort estimation model can be useful for companies that do not have past projects from which to develop their own models. For example, small or immature software companies will not initially have past projects from which to develop their own models [15]. Therefore, the ability to estimate effort using, as a reference, similar projects developed by diverse Web companies is more reasonable than to estimate effort without historical data or previous experience.
- A cross-company data set should only be used for estimation until it is possible for a Web company to build its own within-company model. This is even more appropriate for Web companies that develop Web projects of the same type, using standard technologies and the same staff. Unfortunately it is not possible to know in advance how many projects are needed to construct a useful within company data set since it depends on several factors and specific circumstances

(e.g. the stability of the development process and the number of variables included in the model). However, another study of Web cost estimation within-company data obtained good prediction accuracy using 12 projects [16]. So in some circumstances a dataset of 12 or 13 is sufficient.

- Simple size measures, available early in the lifecycle, that can be estimated by clients (who will almost certainly have no software application knowledge) are suitable for building effort estimation models. Note that to obtain the number of high effort functions it is necessary to provide a client with a list of high effort functions and to count those that have been selected. In addition, the development team size must already be known by the Web company.

Acknowledgements

We would like to thank all those companies that have volunteered data on their Web projects to the Tukutuku database. The Tukutuku database is available to any organization submitting three or more projects to the database.

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Note. The data set can be made available to other researchers for independent audit of the statistical analyses presented in this paper but cannot be published for confidentiality reasons. Please contact Emilia Mendes at emilia@cs.auckland.ac.nz